

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Single Arm, Open Label Phase 2 Study of Tucatinib in Combination With Trastuzumab Deruxtecan in Subjects With Previously Treated Unresectable Locally-Advanced or Metastatic HER2+ Breast Cancer **Short Title/Acronym:** A Study of Tucatinib Plus Trastuzumab Deruxtecan in HER2+ Breast Cancer (HER2CLIMB-04) **Protocol Number:** SGNTUC-025 / NCI-2020-08177 **NCT Number:** NCT04539938 **Posting ID:** 132759827292932301 **Trial Status:** COMPLETED **Sponsor Name:** Pfizer (Seagen is a wholly owned subsidiary of Pfizer) **Company:** Pfizer **Investigational Product:** Tucatinib (Trade Name: Tukysa; ATC Code: L01EH03) in combination with Trastuzumab Deruxtecan (T-DXd) **Phase of Development:** Phase 2 **Medical Monitor:** Pfizer / Seagen Clinical Operations Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, specifically the confirmed Objective Response Rate (cORR), of tucatinib in combination with Trastuzumab Deruxtecan (T-DXd) in subjects with previously treated unresectable locally advanced or metastatic HER2+ breast cancer. **Secondary Objectives:** To assess Progression-Free Survival (PFS), Duration of Response (DOR), Disease Control Rate (DCR), Overall Survival (OS), and the safety/tolerability of the combination regimen.

2.2 Background & Rationale To address the ongoing clinical need for patients with HER2+ metastatic breast cancer (MBC), a disease that remains incurable despite advances in targeted therapies. While T-DXd (an antibody-drug conjugate) has shown significant efficacy in heavily pretreated populations, disease progression eventually occurs. Preclinical xenograft models have shown that adding tucatinib (an oral, highly selective HER2 tyrosine kinase inhibitor) to a HER2-directed ADC increases antitumor activity compared to the ADC alone. The HER2CLIMB-04 study aims to generate clinical evidence to evaluate whether this dual HER2-targeted approach can further improve outcomes, including in patients with active brain metastases.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Parallel

Cohorts: Cohort A without brain metastases; Cohort B with a history of brain metastases)

3.2 Planned Enrollment & Duration Targeted Enrollment: 70 subjects (including a 10-subject safety lead-in phase). **Number of Sites:** Approximately 30 study sites in the United States. **Estimated Study Start:** Late 2020. **Estimated Primary Completion:** Follow-up for duration of response up to approximately 3 years.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, male or female. 2. Confirmed diagnosis of HER2+ breast cancer per ASCO/CAP guidelines (unresectable locally advanced or metastatic). 3. History of prior treatment with a taxane and trastuzumab (with or without pertuzumab) in the locally advanced/metastatic setting OR progression within 6 months after neoadjuvant/adjuvant treatment involving a taxane and trastuzumab. 4. Patients with brain metastases, including active or untreated brain metastases, are permitted.

4.2 Exclusion Criteria 1. Prior treatment with Lapatinib or neratinib within 12 months of starting study treatment (except if given for ≤ 21 days and discontinued for reasons other than disease progression or severe toxicity). 2. Prior treatment with Tucatinib or previously enrolled in a tucatinib clinical trial. 3. Prior treatment with Trastuzumab Deruxtecan (T-DXd) or another antibody-drug conjugate (ADC) consisting of an exatecan derivative. 4. Any systemic anti-cancer therapy ≤ 21 days prior to the first dose of study treatment, or major surgery < 28 days prior.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tucatinib 300 mg twice daily (BID) continuously, combined with Trastuzumab Deruxtecan 5.4 mg/kg administered on Day 1 of each 21-day cycle. **Route of Administration:** Oral (Tucatinib) and Intravenous (T-DXd). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or subject withdrawal. **EMA Product Information:** [Tukysa EPAR](#)

5.2 Concomitant & Prohibited Therapy Permitted: Gonadotropin-releasing hormone (GnRH) agonists used for ovarian suppression in premenopausal women. **Prohibited:** Concurrent investigational agents, any other HER2/EGFR tyrosine kinase inhibitors, or other systemic anti-cancer therapies (including hormonal therapy within the 21-day washout period).

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor/Brain Imaging (MRI/CT), Biomarker Labs | | **Baseline** | Day 1 (Cycle 1) | First dose of Tucatinib and T-DXd, Safety Assessments | | **Interim** | Every 21 Days | Cycle Day 1 visits for IV T-DXd, continuous safety/AE monitoring, PK sampling (C1D12 for lead-in), Efficacy/Tumor scans per RECIST v1.1 | | **End of Study** | Up to 3 Years | Final Vital Signs, Tumor Evaluation, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Confirmed Objective Response Rate (cORR) by investigator assessment. Proportion of subjects with confirmed Complete Response (CR) or Partial Response (PR). **Measurement Method:** Evaluated per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1.

7.2 Secondary & Exploratory Endpoints - Progression-Free Survival (PFS) by investigator assessment per RECIST v1.1. - Duration of Response (DOR) by investigator assessment. - Disease Control Rate (DCR) by investigator assessment. - Overall Survival (OS) and safety profile. - Exploratory endpoints: cORR, PFS, and DOR by Independent Central Review (ICR), pharmacokinetic analyses, and patient-reported outcomes (EQ-5D-5L).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of toxicities (graded via standard oncology toxicity scales), with specific attention paid to hematological disorders, gastrointestinal issues, and Interstitial Lung Disease (ILD)/pneumonitis known to be associated with T-DXd. **Serious Adverse Events (SAEs):** Safety Lead-in cohort (first 10 subjects) followed for at least 1 cycle prior to expanding enrollment to ensure regimen tolerability. **Data Monitoring Committee (DMC):** A Safety Monitoring Committee (SMC) will continuously review safety data, SAEs, dose modifications, and deaths.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy evaluations will be conducted on response-evaluable subjects; safety evaluations on all subjects who receive at least one dose of the investigational product. Cohorts (A and B) will be analyzed both independently and pooled. **Sample Size Justification:** A targeted enrollment of 70 subjects (evenly distributed between patients with and without brain

metastases) is planned to generate descriptive statistics for efficacy and safety. **Statistical Methods:** For response rates (cORR), the 2-sided exact confidence interval will be calculated using the Clopper-Pearson method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Safety lead-in enrollment pausing after 10 subjects for comprehensive SMC safety review before subsequent expansion. **Audit Trail:** Centralized independent review for exploratory efficacy endpoints; electronic Data Capture (eDC) via Pfizer/Seagen data management protocols.

11. Study Identifiers & Governance

Primary ID: SGNTUC-025 **Posting ID:** 132759827292932301 **Registry Number:** NCT04539938 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study of Tucatinib Plus Trastuzumab Deruxtecan in HER2+ Breast Cancer **Official Regulatory Title:** A Single Arm, Open Label Phase 2 Study of Tucatinib in Combination With Trastuzumab Deruxtecan in Subjects With Previously Treated Unresectable Locally-Advanced or Metastatic HER2+ Breast Cancer

12. Clinical Framework (Oncology Specific)

Disease Area: HER2 Positive Breast Cancer **Preferred UMLS Name:** HER2 Positive Breast Neoplasms **Primary Condition:** Unresectable locally advanced or metastatic HER2+ Breast Cancer (with or without brain metastases). **Diagnosis Threshold:** Confirmed HER2+ per ASCO/CAP CLIA-certified laboratory guidelines. **Medical Methodology:** Dual targeted anti-HER2 therapy utilizing a small molecule TKI alongside an Antibody-Drug Conjugate (ADC). **Optimization Target:** Disease control, Objective Response, and intracranial/extracranial tumor reduction.

13. Study Procedures & Methodology

Management Pathway: Standard oncological administration involving 21-day cycles. **Education Protocol (HCP):** Investigator training on recognizing and managing ADC-related ILD/pneumonitis. **Quality Audit Mechanism:** Safety Monitoring Committee (SMC) interim safety gate after the initial 10-patient lead-in cohort.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 3 years of follow-up for duration of response and survival metrics. **Onsite Visit Cadence:** Every 21 days (Cycle 1, Day 1 layout). **Radiographic Assessment Frequency:** Scans performed sequentially to assess progression per RECIST v1.1 criteria.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				